# The question

When contacts-v1-exp199-1.5B emits a contact set in one rollout, is that set jointly realizable as a single 3D structure — more so than a contact set carrying the same per-pair marginals but assembled across different rollouts?

Plainly: does autoregressive generation produce a coherent structural hypothesis, or a bag of independently-drawn contacts that happen to share one document?

Issue #211 — github.com/Open-Athena/MarinFold/issues/211

# Why it is worth asking

Every contacts-v1 eval we have (#82, #89, #180) collapses the 100 rollouts per protein into per-pair vote counts and scores the marginals. The per-rollout partition is discarded at the moment of the increment, so nothing has ever measured whether a single rollout is internally coherent.

Two prior results say the joint is where the information is, and neither tests generation. #201 Phase 0: 77% of the val loss is nuisance permutation entropy, so next-token CE barely scores the joint. #163: conditioning on TRUE partial contact maps lifts R-precision 0.145 to 0.556.

If consistency ranks rollouts without ground truth it is also a best-of-N selector and a candidate RL reward for #200 / #208 — usable on sequences with no known structure, which is the entire ESM-Atlas half of #199's training mixture.

# The method that did not work

The natural approach — Floyd-Warshall bound smoothing over the distances implied by the contacts plus the 3.8 A CA(i)-CA(i+1) bond — has no discriminative power. It reports zero violations for a true contact set AND zero for a separation-matched random one, at four different bound pairs.

The reason is structural, not a tuning failure. A violation needs a path of upper bounds summing below some lower bound. Any two-hop contact path is 2U, about 20-24 A, and a contact plus k backbone steps is U + 3.8k — so nothing reaches under a ~10 A lower bound once min_seq_separation = 6 has excluded the close-in-sequence pairs.

Triangle smoothing tests feasibility in an arbitrary METRIC SPACE, which a contact graph satisfies almost for free. It does not test feasibility in R^3.

# What works instead

The 3D embedding residual: minimise bond, contact, non-contact and steric violation over x in R^(Lx3), and report the leftover contact excess, minimised over restarts.

Floyd-Warshall is kept as step 1 of 3 — it is the bound-smoothing step of the classic Crippen-Havel EMBED algorithm and the right preconditioner for the embedding — but it is not the metric.

Bounds are measured on all 554 ground-truth proteins, not assumed (91,525 contact pairs, 11.1M non-contact pairs), reusing exp174's published GT bundle:

bond = 3.809 A (median CA(i)-CA(i+1))

u_contact = 13.02 A (contact CA-CA p99.5)

l_noncontact = 6.46 A (non-contact CA-CA p0.5)

d_min = 3.53 A (closest non-bonded CA pair, p1)

# The number that governs every claim

10.7% of non-contact pairs sit closer than u_contact.

pyconfind contacts are SIDE-CHAIN contacts, so CA-CA distance is only a proxy for them and no threshold pair separates the two populations. A nonzero residual therefore means "less geometrically consistent", never "provably unrealizable".

The arm comparison is unaffected: every arm is scored under identical bounds, so the scale cancels. But the absolute number is not a proof of anything, and the writeup does not treat it as one.

# The calibration gate passed, and corrected the issue

Ground truth beats a separation-matched random set of the same size and the same |i-j| profile by 5.6x in median per-contact excess, on 89.6% of the 470 chain-break-free proteins — 95.4% at L 100-200, 88.3% at L 200-350, and 100% at L >= 350.

Two of the three criteria written into the issue were wrong. "GT scores about 0" is incoherent against the bounds it was paired with: u_contact is a p99.5, so ~0.5% of real contacts exceed it by construction and ground truth carries a structural nonzero floor. The gate is now relative.

The decoy arm is not a floor. A different real protein's contact map scores the same as the true one (0.0384 vs 0.0337; truth wins on 49.6%, a coin flip). That is correct — the score is sequence-blind — but it bounds the claim: this experiment detects "not a fold at all", not "wrong fold copied from a real one".

Scope limit: below L = 100 the metric is near-blind (GT 0.0000 vs random 0.0011) — a short chain embeds almost anything. A further 15% of the eval set has a chain break and is not embeddable as the continuous chain a contacts-v1 document asserts. Both subsets are reported separately, never silently dropped.

# How sensitive is it where the model actually lives?

The gate contrast is easy. The real contrast is a rollout against a chimera built from the same rollouts with the same marginals — both roughly 60/40 true/false mixtures of the same contacts. Sweeping corruption across #199's operating band (R-precision about 0.59, so about 40% wrong), over 60 proteins at L >= 100:

a 0.05 step in corruption: sign consistency 53-65%, p 0.04-0.35 — NOT reliably separable at n = 60.

a 0.10 step: sign consistency 58.3%, p = 0.006 — separable.

a 0.20 step: sign consistency 73.3%, p = 2.3e-05 — strongly separable.

So the resolvable effect is bounded and the experiment's power has to come from scale. This sweep is also an UPPER bound on the real effect: the chimera keeps every pair the model proposed and only breaks their co-occurrence, a gentler perturbation than swapping pairs for random ones.

# The arms

Every arm is size-matched. #142 measured rollouts emitting about 0.70x the ground-truth contact count, and a sparser set is trivially easier to embed, so an unmatched comparison would read under-generation as consistency.

ground truth — calibration ceiling. GT size-matched — removes the count confound.

rollout — the treatment.

marginal-matched chimera — THE KEY NULL: pairs drawn from the pooled rollout votes with probability proportional to vote count.

splice chimera — half of rollout a plus half of rollout b. Decoy protein — sequence-blindness ceiling. Separation-matched random — floor.

Why the marginal chimera is the sharp test: it shares the model, the protein, the per-pair marginals and the set size with the rollout. The only difference is whether the contacts were drawn jointly in one autoregressive pass or independently from the pooled vote distribution. Any gap is joint structure the model put there while generating, and cannot be explained by marginal accuracy — which is all any existing eval measures.

# Result: yes, and the effect is large

554 proteins x 100 rollouts, 30 replicates per arm, 51,890 scored contact sets. Headline on the 394 chain-break-free proteins at L >= 100.

Rollout vs marginal-matched chimera: mean delta +0.0655 per contact, 95% CI [+0.0561, +0.0752], rollout more consistent on 89.8% of proteins, Wilcoxon p = 1.4e-58.

Rollout vs splice chimera: +0.0339, rollout better on 93.1%, p = 2.4e-62.

For scale: the entire ground-truth-to-random range of this metric is 0.064 to 0.272. The rollout-vs-chimera effect is 0.0655, about 31% of that full range. This is not a marginal result.

A rollout is about as 3D-consistent as the real contact map (0.0562 against 0.0639 for ground truth, 0.0630 for size-matched ground truth), while the chimera built from the same contacts with the same marginals sits at 0.1216.

# The effect grows with protein length

L 100-200: delta +0.0473, rollout better on 88.3% of 240 proteins.

L 200-350: delta +0.0771, rollout better on 90.6% of 128 proteins.

L 350-761: delta +0.1763, rollout better on 100.0% of 26 proteins.

That is the direction #180 predicts, and it is the opposite of an artifact: if the effect were driven by the metric's slack it would shrink as constraints multiply.

Below L = 100 the effect is +0.0029 (70.4%) — the gate already established the metric is near-blind there, and it duly reports nothing. The 84 chain-break proteins, excluded from the headline, show +0.0915 on 95.2%.

# But consistency does NOT rank rollouts by accuracy

The secondary question was whether this buys a reference-free selector. It does not, or barely.

Spearman rho(excess, precision) within a protein: mean -0.0175, useful on 51.8% of proteins — a coin flip.

Picking the most-consistent of 30 rollouts gains +0.0110 precision (95% CI [+0.0032, +0.0188]) against an oracle headroom of +0.1299. That is about 8% of the available headroom: statistically nonzero, practically weak.

The calibration gate already explained why. A decoy protein's contact map scores the same as the truth (0.0668 vs 0.0639) because the score is sequence-blind. It cannot tell a coherent WRONG fold from a coherent RIGHT one — and a rollout can be highly self-consistent and still wrong.

# Conclusion

The model generates a coherent structural hypothesis, not a bag of independently drawn contacts. Sampling the same contacts with the same per-pair marginals but independently costs 0.0655 per contact, on 89.8% of proteins, and the gap widens with length. Autoregressive generation is doing real joint work that every marginals-only eval (#82, #89, #180) discards at the vote-counting step.

The coherence is not accuracy-aligned. Self-consistency is nearly uncorrelated with whether a rollout is right, so it is not a best-of-N selector and, on its own, not an RL reward — it would reinforce coherence the model already has. For #200 / #208 that is the useful negative: pair it with an accuracy signal or skip it.

Read together with #163 — where conditioning on TRUE partial contact maps lifts R-precision 0.145 to 0.556 — the picture is that the model's remaining gap is about CORRECTNESS, not COHERENCE. It already commits to a single self-consistent structure; it commits to the wrong one.

# Caveats, stated plainly

The score is statistical, not a proof. 10.7% of real non-contact pairs sit closer than the contact upper bound, so a nonzero residual means "less geometrically consistent", never "provably unrealizable".

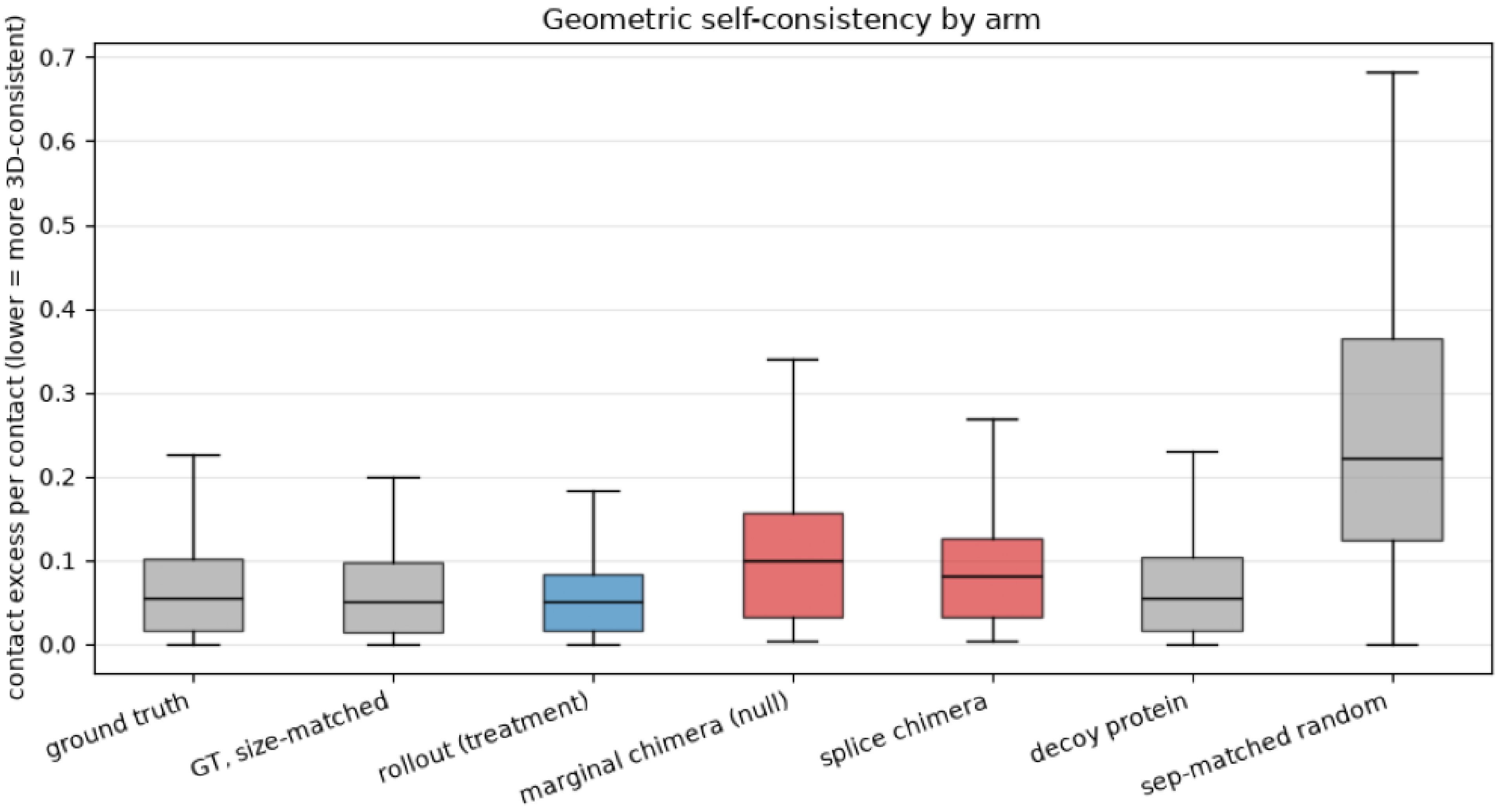
The rollout-beats-ground-truth comparison is confounded and is not claimed. The model preferentially emits short-range contacts (precision 0.679 short vs 0.566 long), and short-range contacts are geometrically easier to satisfy. The primary contrast is NOT confounded this way: the chimera is drawn from the same pooled rollout contacts with the same marginals, so it carries the same separation profile.

Below L = 100 the metric is uninformative, and 15% of the eval set has a chain break. Both subsets are reported separately rather than dropped.

Cost: 95.8 min to generate 55,400 rollouts and 494 min to score 51,890 contact sets, both on one RTX A5000. The CoreWeave fan-out was written but unusable — the workstation's credentials are dead.

# arms.png

Per-protein mean consistency by arm. Blue = the rollout; red = the size- and marginal-matched nulls.




```
$ plot_results.py --scores data/arm_scores.csv
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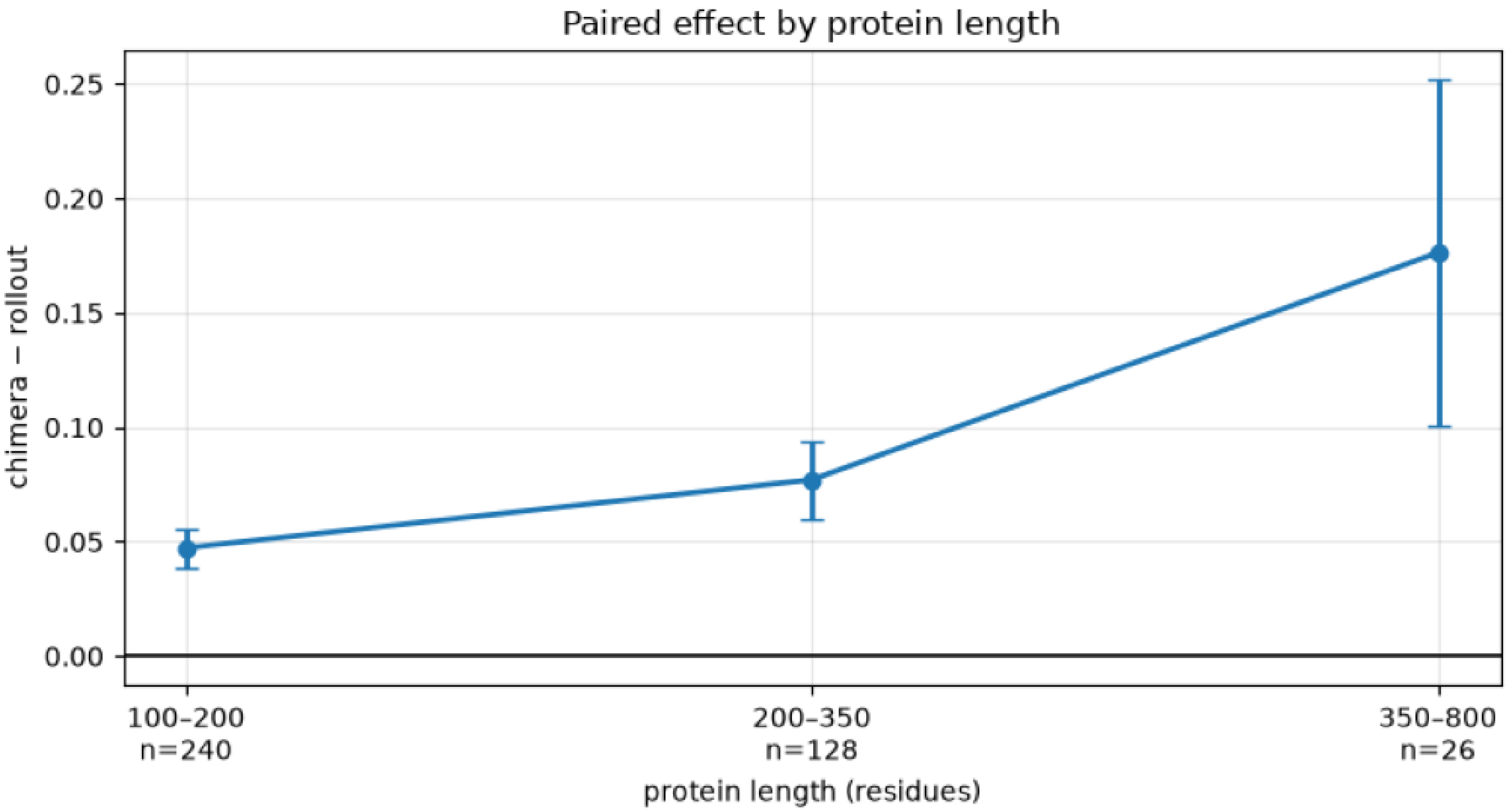
## by_length.png

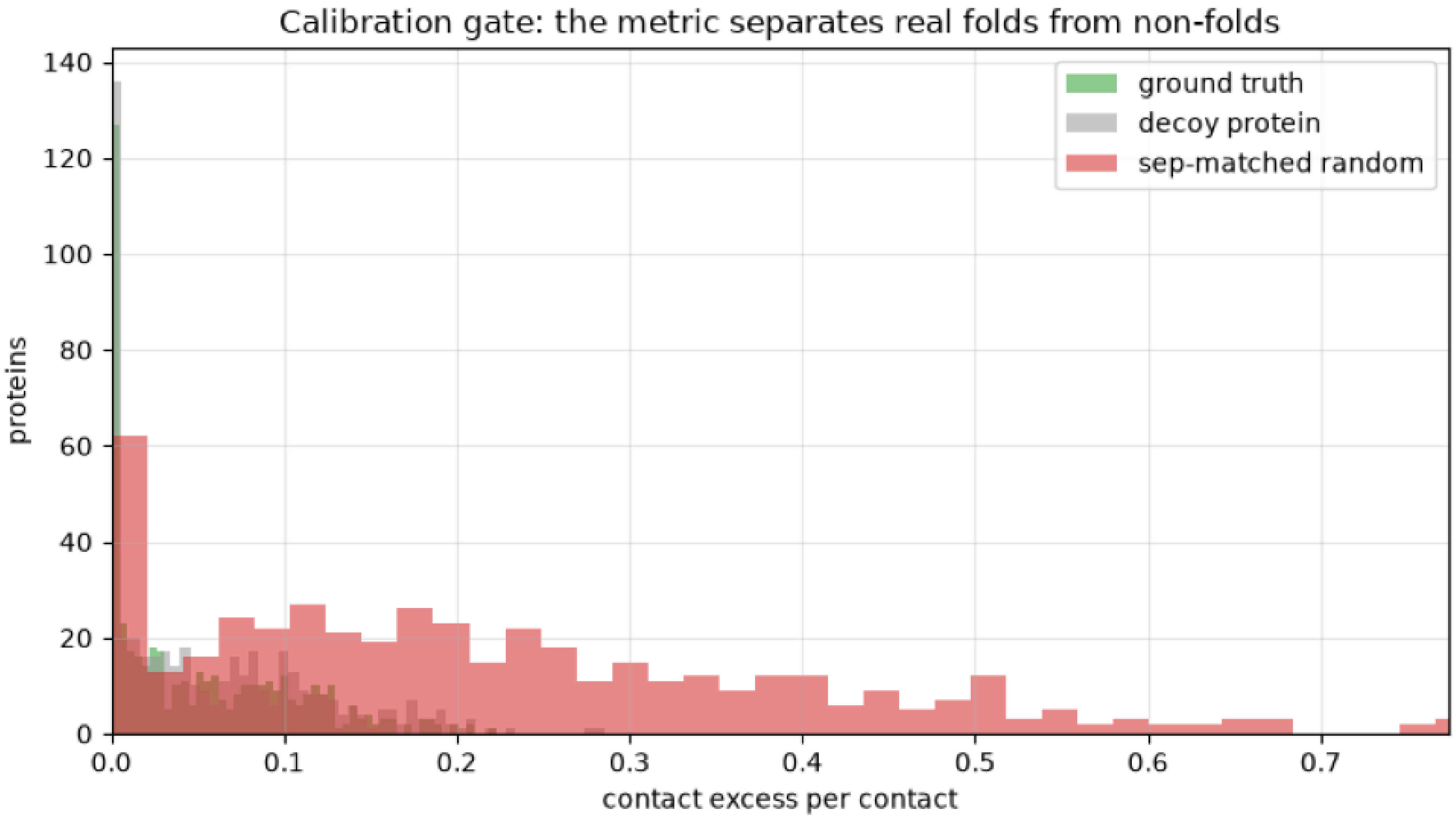
Paired effect stratified by length, 95% CI. The metric is near-blind below L≈100, so that bin is excluded.




`$ plot_results.py --scores data/arm_scores.csv`

# calibration_gate.png

Ground truth sits 5.6× below random. A decoy protein ties with the truth — the score is sequence-blind.

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$ plot_results.py --scores data/arm_scores.csv
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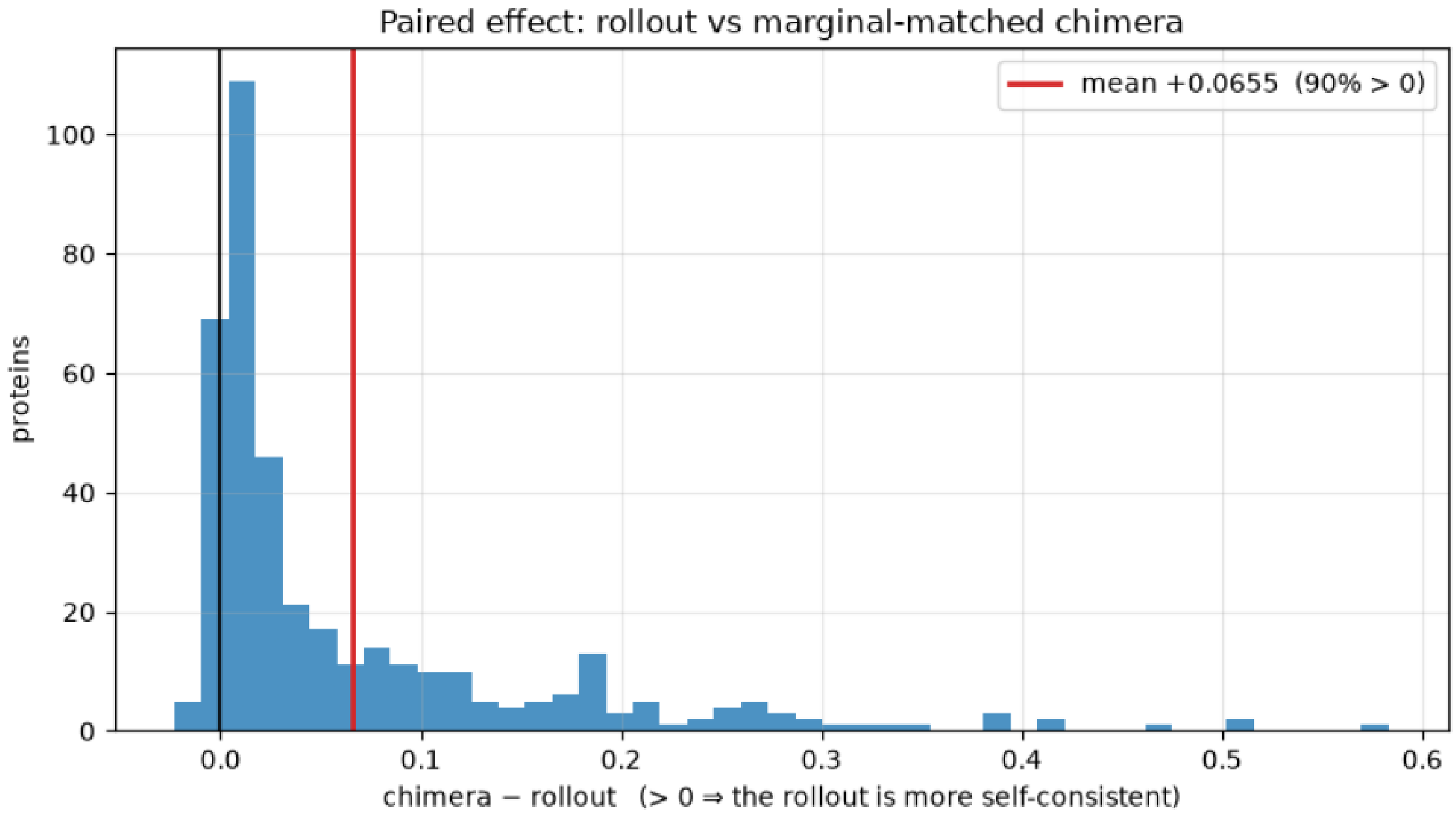
# dose_response.png

Score vs corruption, 95% CI. Grey band marks where #199 actually generates (R-precision ≈ 0.59).


`$ plot_results.py --scores data/arm_scores.csv`

## paired_effect.png

Per-protein paired difference. Positive means the jointly-generated set embeds better than its marginal-matched null.

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$ plot_results.py --scores data/arm_scores.csv
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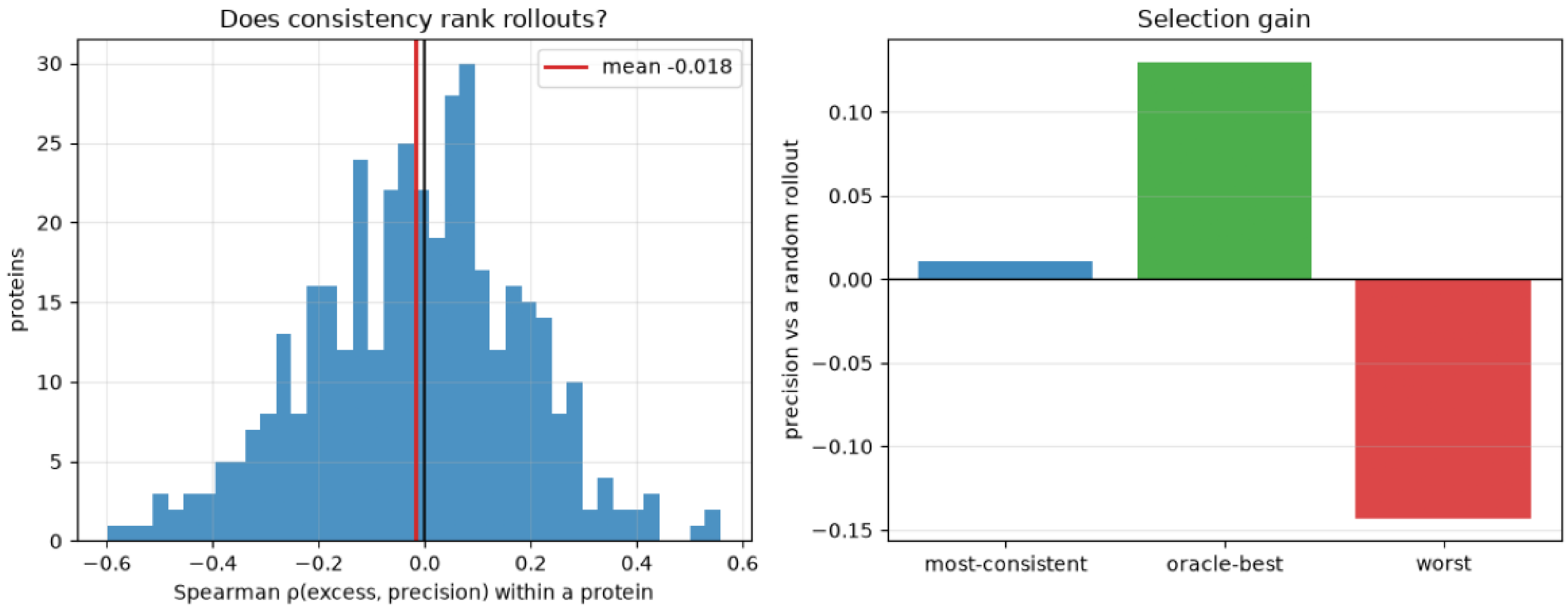
# selection.png

Left: per-protein rank correlation (negative = useful). Right: precision gained by selecting on consistency, against the oracle headroom.




`$ plot_results.py --scores data/arm_scores.csv`